Online appendix to: A new approach for open-end sequential change point monitoring

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July 25, 2020

A Proofs of the results in Section 2

Proof of Theorem 2.7: In the proof we use the following extra notation. Define the statistic

$$E_m(k) = m^{-1/2} \max_{j=0}^{k-1} (k-j) \left\| \hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k} \right\|_{\Sigma^{-1}} , \tag{A.1}$$

where we have replaced the long-run variance estimator $\hat{\Sigma}_m$ by the (unknown) true long-run variance Σ in definition (2.2). Further define

$$\tilde{E}_m(k) = m^{-1/2} \max_{j=0}^{k-1} (k-j) \left\| \frac{1}{m+j} \sum_{t=1}^{m+j} \mathcal{IF}_t - \frac{1}{k-j} \sum_{t=m+j+1}^{m+k} \mathcal{IF}_t \right\|_{\Sigma^{-1}} , \tag{A.2}$$

where we have replaced the estimator $\hat{\theta}_i^j$ by corresponding averages of the influence function in (A.1). Throughout the proof we will frequently use that due to continuity of $\tilde{w}$

and $\limsup_{t\to\infty} t\tilde{w}(t) < \infty$ in Assumption 2.4, the weight function w has a uniform upper bound, say u_w. Finally, the triple $(\Omega, \mathcal{A}, \mathbb{P})$ will denote the underlying probability space.

The proof itself is now split into several Lemmas A.1 - A.5. The first Lemma shows that $\tilde{E}_m(k)$ and $E_m(k)$ are (asymptotically) equivalent. Lemma A.2 will approximate $E_m(k)$ by Brownian motions, while Lemma A.3 then yields a limit for this approximation. Lemma A.4 finishes the proof by plugging in the covariance estimator, meaning that $\hat{E}_m(k)$ and $E_m(k)$ are asymptotically equivalent. Finally, Lemma A.5 will establish the other representation of the limit distribution in (2.12). In each Lemma, we suppose that the assumptions of Theorem 2.7 are valid.

Lemma A.1 (Remove Remainders) *It holds that*

$$\sup_{k=1}^{\infty} w(k/m)\Big|E_m(k) - \tilde{E}_m(k)\Big| = o_{\mathbb{P}}(1)$$

as $m \to \infty$.

Proof. By the (reverse) triangle inequality and the linearization in (2.7) we obtain

$$\begin{aligned}
&\Big|E_m(k) - \tilde{E}_m(k)\Big| \\
&\leq m^{-1/2} \max_{j=0}^{k-1}(k-j)\Big\|\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k} - \frac{1}{m+j}\sum_{t=1}^{m+j}\mathcal{IF}_t + \frac{1}{k-j}\sum_{t=m+j+1}^{m+k}\mathcal{IF}_t\Big\|_{\Sigma^{-1}} \\
&= m^{-1/2} \max_{j=0}^{k-1}(k-j)\Big\|R_{1,m+j} - R_{m+j+1,m+k}\Big\|_{\Sigma^{-1}} ,
\end{aligned}$$

where we used that θ_t is constant for the last equality. Next, we obtain that

$$\begin{aligned}
&\sup_{k=1}^{\infty} \frac{w(k/m)}{m^{1/2}} \max_{j=0}^{k-1}(k-j)\Big\|R_{m+j+1,m+k}\Big\|_{\Sigma^{-1}} \\
=&\sup_{k=1}^{\infty} \frac{w(k/m)(m+k)^{1/2}}{m^{1/2}} \max_{j=0}^{k-1} \frac{k-j}{(m+k)^{1/2}}\Big\|R_{m+j+1,m+k}\Big\|_{\Sigma^{-1}} \\
\leq&\sup_{k=1}^{\infty} \frac{w(k/m)(m+k)^{1/2}}{m^{1/2}} \sup_{k=1}^{\infty} \max_{1\leq i<j\leq m+k} \frac{j-i+1}{(m+k)^{1/2}}\Big\|R_{i,j}\Big\|_{\Sigma^{-1}} .
\end{aligned} \tag{A.3}$$

Similar as in (A.3) it holds

$$\begin{aligned}
&\sup_{k=1}^{\infty} \frac{w(k/m)}{m^{1/2}} \max_{j=0}^{k-1}(k-j)\Big\|R_{1,m+j}\Big\|_{\Sigma^{-1}} \\
\leq&\sup_{k=1}^{\infty} \frac{w(k/m)(m+k)}{m}\sqrt{m}\sup_{k=1}^{\infty}\max_{j=0}^{k-1}\Big\|R_{1,m+j}\Big\|_{\Sigma^{-1}} \\
=&\sup_{k=1}^{\infty} \frac{w(k/m)(m+k)}{m}\sup_{k=0}^{\infty}\sqrt{m}\Big\|R_{1,m+k}\Big\|_{\Sigma^{-1}}
\end{aligned} \tag{A.4}$$

Using Assumption 2.4 for the weight function w we obtain

$$\sup_{k=1}^{\infty} \frac{w(k/m)(m+k)}{m} = \sup_{k=1}^{\infty} w(k/m)\Big(1+\frac{k}{m}\Big) \leq \sup_{t>0} w(t)(1+t) < \infty \ . \tag{A.5}$$

and by similar arguments it holds that

$$\sup_{k=1}^{\infty} \frac{(m+k)^{1/2} w(k/m)}{\sqrt{m}} < \infty \ . \tag{A.6}$$

Further note that due to Assumption 2.5 with probability one

$$\begin{aligned} \sup_{k=1}^{\infty} \sqrt{m}\Big|R_{1,m+k}\Big| \leq \sup_{k=1}^{\infty} \sqrt{m+k}\Big|R_{1,m+k}\Big| &= \sup_{k=m}^{\infty} \sqrt{k}\Big|R_{1,k}\Big| \\ &\overset{j=k,i=1}{\leq} \sup_{k=m}^{\infty} \max_{1\leq i<j\leq k} \frac{j-i+1}{\sqrt{k}}\Big|R_{i,j}\Big| = o(1) \ . \end{aligned} \tag{A.7}$$

Now combining (A.5), (A.6) and (A.7) the bounds derived in (A.3) and (A.4) are of order $o_{\mathbb{P}}(1)$, which finishes the proof of Lemma A.1. □

For the proof of the next Lemma we can proceed (roughly) similar to the proof of Lemma 5.2 in Fremdt (2015).

Lemma A.2 (Approximation with Brownian motions) *Define*

$$P_m(k) = \frac{1}{\sqrt{m}} \max_{j=0}^{k-1} \Big|W_{m,1}(k) - \frac{m+k}{m+j} W_{m,1}(j) + \frac{k-j}{m+j} W_{m,2}(m)\Big| \ ,$$

then

$$\sup_{k=1}^{\infty} w(k/m)\Big|\tilde{E}_m(k) - P_m(k)\Big| = o_{\mathbb{P}}(1)$$

as $m \to \infty$.

Proof. For the remainder of the proof let $W_{m,i}^{\Sigma} := \sqrt{\Sigma} W_{m,i}$ for $i = 1, 2$ and note that this implies

$$P_m(k) = \frac{1}{\sqrt{m}} \max_{j=0}^{k-1} \Big\|W_{m,1}^{\Sigma}(k) - \frac{m+k}{m+j} W_{m,1}^{\Sigma}(j) + \frac{k-j}{m+j} W_{m,2}^{\Sigma}(m)\Big\|_{\Sigma^{-1}} \ .$$

The last display and the (reverse) triangle inequality then yield

$$
\begin{aligned}
&\left|\tilde{E}_m(k) - P_m(k)\right| \\
&\leq \frac{1}{\sqrt{m}} \max_{j=0}^{k-1} \Big\| \frac{k-j}{m+j} \sum_{t=1}^{m+j} \mathcal{IF}_t - \sum_{t=m+j+1}^{m+k} \mathcal{IF}_t \\
&\qquad + W_{m,1}^{\Sigma}(k) - \frac{m+k}{m+j} W_{m,1}^{\Sigma}(j) + \frac{k-j}{m+j} W_{m,2}^{\Sigma}(m) \Big\|_{\Sigma^{-1}} \qquad \text{(A.8)} \\
&\leq \frac{1}{\sqrt{m}} \Big\| \sum_{t=m+1}^{m+k} \mathcal{IF}_t - W_{m,1}^{\Sigma}(k) \Big\|_{\Sigma^{-1}} + \frac{1}{\sqrt{m}} \max_{j=0}^{k-1} \frac{m+k}{m+j} \Big\| \sum_{t=m+1}^{m+j} \mathcal{IF}_t - W_{m,1}^{\Sigma}(j) \Big\|_{\Sigma^{-1}} \\
&\qquad + \frac{1}{\sqrt{m}} \max_{j=0}^{k-1} \frac{k-j}{m+j} \Big\| \sum_{t=1}^{m} \mathcal{IF}_t - W_{m,2}^{\Sigma}(m) \Big\|_{\Sigma^{-1}} .
\end{aligned}
$$

We will treat the three summands of the last display separately. Using the definition of the operator norm, we derive the following bound for the first summand

$$
\begin{aligned}
&\sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \Big\| \sum_{t=m+1}^{m+k} \mathcal{IF}_t - W_{m,1}^{\Sigma}(k) \Big\|_{\Sigma^{-1}} \\
&\leq \sup_{k=1}^{\infty} \frac{w(k/m)k^{\xi}}{\sqrt{m}} \sup_{k=1}^{\infty} \frac{1}{k^{\xi}} \Big\| \sum_{t=m+1}^{m+k} \mathcal{IF}_t - W_{m,1}^{\Sigma}(k) \Big\|_{\Sigma^{-1}}
\end{aligned}
$$

By Assumption 2.3 and the estimate $\|x\|_{\Sigma^{-1}} \leq \|\Sigma^{-1}\|_{op}^{1/2}|x|$ for all $x \in \mathbb{R}^p$, the second factor is of order $\mathcal{O}_{\mathbb{P}}(1)$. Since w has an upper bound u_w we obtain for the first factor, that

$$
\begin{aligned}
\sup_{k=1}^{m} \frac{w(k/m)k^{\xi}}{\sqrt{m}} &\leq \sup_{k=1}^{m} \frac{u_w k^{\xi}}{\sqrt{m}} = u_w m^{\xi-1/2} = o(1) , \\
\sup_{k=m+1}^{\infty} \frac{w(k/m)k^{\xi}}{\sqrt{m}} &\leq \sup_{k=m+1}^{\infty} \frac{1}{k^{1/2-\xi}} \sup_{k=1}^{\infty} \frac{w(k/m)k^{1/2}}{\sqrt{m}} \leq \frac{1}{m^{1/2-\xi}} \sup_{t>0} \sqrt{t} w(t) = o(1) .
\end{aligned}
$$

Next, we can bound the second summand on the right-hand side in (A.8) by

$$
\begin{aligned}
&\sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \max_{j=0}^{k-1} \frac{m+k}{m+j} \Big\| \sum_{t=m+1}^{m+j} \mathcal{IF}_t - W_{m,1}^{\Sigma}(j) \Big\|_{\Sigma^{-1}} \\
&\leq \sup_{k=1}^{\infty} \frac{w(k/m)(m+k)}{\sqrt{m}} \max_{j=1}^{k-1} \frac{1}{j^{\xi} m^{1-\xi}} \Big\| \sum_{t=m+1}^{m+j} \mathcal{IF}_t - W_{m,1}^{\Sigma}(j) \Big\|_{\Sigma^{-1}} \\
&\leq \sup_{k=1}^{\infty} \frac{w(k/m)(m+k)m^{\xi-1}}{\sqrt{m}} \sup_{k=1}^{\infty} \frac{1}{k^{\xi}} \Big\| \sum_{t=m+1}^{m+k} \mathcal{IF}_t - W_{m,1}^{\Sigma}(k) \Big\|_{\Sigma^{-1}} ,
\end{aligned}
$$

where we used that $(m+j) = (m+j)^{\xi}(m+j)^{1-\xi} \geq j^{\xi} m^{1-\xi}$. Using again Assumption 2.3,

the second factor in the last display is of order $\mathcal{O}_{\mathbb{P}}(1)$. Moreover, following the idea of the proof of Lemma 3 in Aue et al. (2006) it holds that

$$\begin{aligned}
\sup_{k=1}^{m} \frac{w(k/m)(m+k)m^{\xi-1}}{\sqrt{m}} &\leq u_w \sup_{k=1}^{m}(m+k)m^{\xi-3/2} = u_w 2m^{\xi-1/2} = o(1) \ , \\
\sup_{k=m+1}^{\infty} \frac{w(k/m)(m+k)m^{\xi-1}}{\sqrt{m}} &= m^{\xi-1/2} \sup_{k=m+1}^{\infty} \big(1+k/m\big)w(k/m) \\
&\leq m^{\xi-1/2} \sup_{t>1}(1+t)w(t) = o(1)
\end{aligned} \tag{A.9}$$

and it remains to treat the third summand of the right-hand side in (A.8), which can be bounded by

$$\begin{aligned}
&\sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \max_{j=0}^{k-1} \frac{k-j}{m+j} \Big\| \sum_{t=1}^{m} \mathcal{IF}_t - W_{m,2}^{\Sigma}(m) \Big\|_{\Sigma^{-1}} \\
\leq &\sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \frac{k}{m} \Big\| \sum_{t=1}^{m} \mathcal{IF}_t - W_{m,2}^{\Sigma}(m) \Big\|_{\Sigma^{-1}} \\
\leq &\sup_{k=1}^{\infty} \frac{w(k/m)km^{\xi-1}}{\sqrt{m}} \frac{1}{m^{\xi}} \Big\| \sum_{t=1}^{m} \mathcal{IF}_t - W_{m,2}^{\Sigma}(m) \Big\|_{\Sigma^{-1}} \ .
\end{aligned}$$

Using Assumption 2.3 and the arguments in (A.9) this term is of order $o_{\mathbb{P}}(1)$, which finishes the proof of Lemma A.2. □

Lemma A.3 (Obtain limit process) *The following weak convergence holds*

$$\sup_{k=1}^{\infty} w(k/m)P_m(k) \overset{\mathcal{D}}{\Longrightarrow} \sup_{0\leq t<\infty} \max_{0\leq s\leq t} w(t) \Big| W_1(t) - \frac{1+t}{1+s} W_1(s) + \frac{t-s}{1+s} W_2(1) \Big|$$

as $m \to \infty$, where W_1 and W_2 denote independent, p-dimensional standard Brownian motions.

Proof. First note that due to the scaling properties of the Brownian motion [see for example page 30 of Shorack and Wellner (2009)], it holds in distribution that

$$P_m(k) \overset{\mathcal{D}}{=} \max_{j=0}^{k-1} \Big| W_1(k/m) - \frac{1+k/m}{1+j/m} W_1(j/m) + \frac{k/m - j/m}{1+j/m} W_2(1) \Big| := \widetilde{P}_m(k) \tag{A.10}$$

and so within the proof we will - without loss of generality - only consider $\widetilde{P}_m(k)$. Additionally, we define the processes

$$\begin{aligned}
L^{(1)}(s,t) &:= \tilde{w}(t)\Big(\frac{t+1}{s+1} W_1(s) - W_1(t)\Big) \ , \\
L^{(2)}(s,t) &:= \tilde{w}(t)\frac{t-s}{s+1} W_2(1) \ .
\end{aligned}$$

We will show that $L^{(i)}(s,t)$ for $i=1,2$ are uniformly continuous on $\mathbb{R}^+_\Delta = \{(s,t) \in \mathbb{R}^2 \mid 0 \leq s \leq t \leq T_w\}$ with probability one, where $T_w \in \mathbb{R}_{>0} \cup \{\infty\}$ is the (right) cutoff constant from Assumption 2.4. In case $T_w < \infty$ this directly follows as both processes are already a.s. continuous, so only $T_w = \infty$ is of interest.

Case $L^{(1)}$: In the following let $\varepsilon > 0$ be fixed but arbitrary. Next, we fix one $\omega_0 \in \Omega$, such that W_1 fulfills the law of iterated logarithm and is continuous. As this event has probability one, it suffices to show that $L^{(1)}$ is uniformly continuous for $W_1 = W_1(\cdot, \omega_0)$. For the ease of reading we will omit ω_0 in the presentation below. By the law of iterated logarithm, there exist $C = C(\varepsilon, \omega_0)$ sufficiently large, such that

$$\sup_{t \geq C} \frac{|W_1(t)|}{t} < \frac{\varepsilon}{4B} \ , \tag{A.11}$$

with B chosen as

$$B := \sup_{t>0} (t+2)\tilde{w}(t) < \infty \ .$$

Depending on C, we can split $\mathbb{R}^+_\Delta$ into the (overlapping) sets

$$\mathbb{R}^+_\Delta = \mathcal{M}_1(C) \cup \mathcal{M}_2(C) \cup \mathcal{M}_3(C) \ , \tag{A.12}$$

where we use the definitions

$$\begin{aligned}
\mathcal{M}_1(C) &= \mathbb{R}^+_\Delta \cap [0, C+1]^2 \ , \\
\mathcal{M}_2(C) &= \mathbb{R}^+_\Delta \cap [0, C+1] \times [C, \infty) \ , \\
\mathcal{M}_3(C) &= \mathbb{R}^+_\Delta \cap [C, \infty)^2 \ .
\end{aligned}$$

Further let d denote the maximum distance that is

$$d\big((s_1,t_1),(s_2,t_2)\big) = \max\big\{|s_1 - s_2|, |t_1 - t_2|\big\} \ .$$

Note that by construction of the decomposition in (A.12), whenever $d((s_1,t_1),(s_2,t_2)) < \delta$ for sufficiently small $\delta > 0$, then there is $j \in 1,2,3$, such that both pairs are in the same subset $\mathcal{M}_j(C)$. Thus the uniform continuity of $L^{(1)}$ follows if we can choose $\delta > 0$ sufficiently small, such that

$$\sup_{\substack{d((s_1,t_1),(s_2,t_2))<\delta \\ (s_1,t_1),(s_2,t_2) \in \mathcal{M}_j(C)}} \Big| L^{(1)}(s_1,t_1) - L^{(1)}(s_2,t_2) \Big| < \varepsilon \tag{A.13}$$

for $j = 1,2,3$. In the following, we will treat each subset separately.

Set $\mathcal{M}_1(C)$: As this set is compact, the (ordinary) almost sure continuity of $L^{(1)}$ already implies that (A.13) holds for $j = 1$ and $\delta > 0$ sufficiently small.

Set $\mathcal{M}_2(C)$: We have the following bound

$$\begin{aligned}
&\sup_{\substack{d((s_1,t_1),(s_2,t_2))<\delta \\ (s_1,t_1),(s_2,t_2)\in\mathcal{M}_2(C)}} \left|L^{(1)}(s_1,t_1) - L^{(1)}(s_2,t_2)\right| \\
&\leq \sup_{\substack{|t_1-t_2|<\delta \\ 0\leq s\leq C+1,\ t_1,t_2>C}} \left|L^{(1)}(s,t_1) - L^{(1)}(s,t_2)\right| + \sup_{\substack{|s_1-s_2|<\delta \\ 0\leq s_1,s_2\leq C+1,\ t>C}} \left|L^{(1)}(s_1,t) - L^{(1)}(s_2,t)\right| .
\end{aligned} \tag{A.14}$$

We will treat both summands of the last display individually. The first one can be bounded again as follows

$$\begin{aligned}
&\sup_{\substack{|t_1-t_2|<\delta \\ 0\leq s\leq C+1,\ t_1,t_2>C}} \left|L^{(1)}(s,t_1) - L^{(1)}(s,t_2)\right| \\
&\leq \sup_{\substack{|t_1-t_2|<\delta \\ 0\leq s\leq C+1,\ t_1,t_2>C}} \left|\tilde{w}(t_1) - \tilde{w}(t_2)\right| \left|\frac{t_1+1}{s+1}W_1(s) - W_1(t_1)\right| \\
&\qquad + \tilde{w}(t_2)\left|\frac{t_1-t_2}{s+1}W_1(s) - W_1(t_1) + W_1(t_2)\right|
\end{aligned} \tag{A.15}$$

and again we will treat both terms separately. For the first term note the upper bound

$$\begin{aligned}
&\sup_{\substack{|t_1-t_2|<\delta \\ 0\leq s\leq C+1,\ t_1,t_2>C}} \left|\tilde{w}(t_2) - \tilde{w}(t_1)\right|(t_1+1)\left|\frac{W_1(s)}{s+1} - \frac{W_1(t_1)}{t_1+1}\right| \\
&\qquad \leq 2u_w \sup_{|t_1-t_2|<\delta} \left|\frac{1}{\tilde{w}(t_2)} - \frac{1}{\tilde{w}(t_1)}\right| \sup_{t\geq C}(t+1)\tilde{w}(t) \sup_{t\geq 0}\frac{|W_1(t)|}{t+1} ,
\end{aligned}$$

and since $1/\tilde{w}$ is uniformly continuous this expression is smaller than $\varepsilon/3$ for sufficiently small δ. For the second term of the right-hand side of (A.15), note that we have the bound

$$\begin{aligned}
&\sup_{\substack{|t_1-t_2|<\delta \\ 0\leq s\leq C+1,\ t_1,t_2>C}} \tilde{w}(t_2)\left|\frac{t_1-t_2}{s+1}W_1(s) - W_1(t_1) + W_1(t_2)\right| \\
&\leq \sup_{\substack{|t_1-t_2|<\delta \\ 0\leq s\leq C+1,\ t_1,t_2>C}} \tilde{w}(t_2)|t_1-t_2|\frac{|W_1(s)|}{s+1} + \tilde{w}(t_2)(t_1+1)\frac{|W_1(t_1)|}{t_1+1} + \tilde{w}(t_2)(t_2+1)\frac{|W_1(t_2)|}{t_2+1} \\
&\leq \delta u_w \sup_{s\geq 0}\frac{|W_1(s)|}{s+1} + 2\sup_{t>0}(t+2)\tilde{w}(t)\cdot\sup_{t>C}\frac{|W_1(t)|}{t+1}
\end{aligned}$$

and by the choice of C in (A.11) and for sufficiently small δ this is bounded by $\varepsilon/3$. To complete the treatment of $\mathcal{M}_2(C)$ it only remains to examine the second term on the

right-hand side of (A.14). We obtain that

$$\sup_{\substack{|s_1-s_2|<\delta \\ 0\le s_1,s_2\le C+1,\ t>C}} \left|L^{(1)}(s_1,t)-L^{(1)}(s_2,t)\right|$$
$$\le \sup_{t>0}(t+1)\tilde{w}(t) \sup_{\substack{|s_1-s_2|<\delta \\ 0\le s_1,s_2\le C+1}} \left|\frac{W_1(s_1)}{s_1+1}-\frac{W_1(s_2)}{s_2+1}\right| ,$$

which can be bounded by $\varepsilon/3$ for sufficiently small δ since the first factor is a constant and the function $f(s)=W_1(s+1)/(s+1)$ is uniformly continuous on the compact set $[0,C+1]$.

Set $\mathcal{M}_3(C)$: Note that

$$\sup_{\substack{d((s_1,t_1),(s_2,t_2))<\delta \\ (s_1,t_1),(s_2,t_2)\in\mathcal{M}_3(C)}} \left|L^{(1)}(s_1,t_1)-L^{(1)}(s_2,t_2)\right| \le 2\sup_{s,t>C}|L^{(1)}(s,t)|$$
$$\le 2\sup_{t>0}(t+1)\tilde{w}(t)\sup_{s,t>C}\left|\frac{W_1(s)}{s+1}-\frac{W_1(t)}{t+1}\right|$$
$$\le 4\sup_{t>0}(t+1)\tilde{w}(t)\sup_{t\ge C}\frac{|W_1(t)|}{t}<\varepsilon ,$$

where we used the choice of C in (A.11) for the last estimate.
This completes the third case and so the almost sure uniform continuity of $L^{(1)}$ on the set $\mathbb{R}^+_\Delta$ is established.

Case $L^{(2)}$: Again let $\varepsilon>0$ and suppose that $d\big((s_1,t_1),(s_2,t_2)\big)<\delta$. It holds that

$$\left|L^{(2)}(s_1,t_1)-L^{(2)}(s_2,t_2)\right| \le \left|L^{(2)}(s_1,t_1)-L^{(2)}(s_2,t_1)\right| + \left|L^{(2)}(s_2,t_1)-L^{(2)}(s_2,t_2)\right| \tag{A.16}$$

and note for the first summand of the last display that

$$\left|L^{(2)}(s_1,t_1)-L^{(2)}(s_2,t_1)\right| = |W_2(1)|\tilde{w}(t_1)\left|\frac{t_1-s_1}{s_1+1}-\frac{t_1-s_2}{s_2+1}\right|$$
$$= |W_2(1)|(t_1+1)\tilde{w}(t_1)\left|\frac{s_2-s_1}{(s_1+1)(s_2+1)}\right|$$
$$\le \big|W_2(1)\big|(t_1+1)\tilde{w}(t_1)\big|s_2-s_1\big|$$

and by Assumption 2.4 the last term is smaller than $\varepsilon/2$ uniformly for all $t_1>0$ if $\delta>0$ is chosen sufficiently small. It remains to examine the second summand of the right-hand

side of (A.16). It holds that

$$
\begin{aligned}
|L^{(2)}(s_2,t_1) - L^{(2)}(s_2,t_2)| &\le |\tilde{w}(t_1) - \tilde{w}(t_2)| \frac{|t_1 - s_2|}{s_2+1} |W_2(1)| + \tilde{w}(t_2)|W_2(1)| \frac{|t_2 - t_1|}{s_2+1} \\
&\le \left| \frac{1}{\tilde{w}(t_1)} - \frac{1}{\tilde{w}(t_2)} \right| \tilde{w}(t_1)\tilde{w}(t_2) \frac{t_1+\delta}{s_2+1} |W_2(1)| + u_w |W_2(1)||t_2 - t_1| \\
&\le u_w \left| \frac{1}{\tilde{w}(t_1)} - \frac{1}{\tilde{w}(t_2)} \right| \tilde{w}(t_1)(t_1+\delta)|W_2(1)| + u_w|W_2(1)||t_2 - t_1| \ ,
\end{aligned}
$$

where we used that $s_2 \le t_2 \le t_1 + \delta$ whenever $d\big((s_1,t_1),(s_2,t_2)\big) < \delta$. By Assumption 2.4 the last display is smaller than $\varepsilon/2$ whenever $\delta > 0$ is sufficiently small and so the almost sure continuity of $L^{(2)}$ is shown.

Finally, we can combine our observations to finish the proof. Note that by the results above, also the process $L(s,t) := |L^{(1)}(s,t) + L^{(2)}(s,t)|$ is uniformly continuous with probability one. Next, recall the cutoff parameters from Assumption 2.4 and observe the identity

$$
\sup_{t_w m \le t < T_w m} \sup_{0 \le s \le t} L\big(\lfloor s \rfloor/m, \lfloor t \rfloor/m\big) = \sup_{k=1}^{\infty} w(k/m)\widetilde{P}_m(k) = \sup_{k=t_w \cdot m}^{T_w m} \tilde{w}(k/m)\widetilde{P}_m(k) \ .
$$

Furthermore, note that

$$
\sup_{1 \le t < \infty} \sup_{0 \le s \le t} L\big(\lfloor s \rfloor/m, \lfloor t \rfloor/m\big) = \sup_{0 \le t < \infty} \sup_{0 \le s \le t} L\big(\lfloor s \rfloor/m, \lfloor t \rfloor/m\big) + o(1)
$$

almost surely as $m \to \infty$. Now we can finish the proof of Lemma A.3 using the almost sure uniform continuity of L, which implies that for arbitrary $\varepsilon > 0$ and almost every $\omega \in \Omega$ we can choose sufficiently large $m = m(\varepsilon, \omega)$ such that

$$
\begin{aligned}
&\left| \sup_{k=1}^{\infty} w(k/m)\tilde{P}_m(k) - \sup_{0 \le t < \infty} \max_{0 \le s \le t} w(t) \Big| W_1(t) - \frac{1+t}{1+s} W_1(s) + \frac{t-s}{1+s} W_2(1) \Big| \right| \\
&= \left| \sup_{k=t_w \cdot m}^{T_w m} \tilde{w}(k/m)\tilde{P}_m(k) - \sup_{t_w \le t < T_w} \max_{0 \le s \le t} \tilde{w}(t) \Big| W_1(t) - \frac{1+t}{1+s} W_1(s) + \frac{t-s}{1+s} W_2(1) \Big| \right| \\
&= \left| \sup_{t_w m \le t < T_w m} \sup_{0 \le s \le t} L\big(\lfloor s \rfloor/m, \lfloor t \rfloor/m\big) - \sup_{t_w \le t < T_w} \sup_{0 \le s \le t} L\big(s,t\big) \right| \\
&= \left| \sup_{t_w m \le t < T_w m} \sup_{0 \le s \le t} L\big(\lfloor s \rfloor/m, \lfloor t \rfloor/m\big) - \sup_{t_w m \le t < T_w m} \sup_{0 \le s \le t} L\big(s/m, t/m\big) \right| \\
&\le \sup_{t_w m \le t < T_w m} \sup_{0 \le s \le t} \Big| L\big(\lfloor s \rfloor/m, \lfloor t \rfloor/m\big) - L\big(s/m, t/m\big) \Big| \\
&\le \sup_{0 \le t < \infty} \sup_{0 \le s \le t} \Big| L\big(\lfloor s \rfloor/m, \lfloor t \rfloor/m\big) - L\big(s/m, t/m\big) \Big| \\
&\le \sup_{\substack{d((s_1,t_1),(s_2,t_2)) < 1/m \\ (s_1,t_1),(s_2,t_2) \in \mathbb{R}^+_\Delta}} \Big| L\big(s_1,t_1\big) - L\big(s_2,t_2\big) \Big| < \varepsilon \ .
\end{aligned}
$$

□

Combining Lemma A.1, A.2 and A.3 we have already proven that

$$\sup_{k=1}^{\infty} w(k/m)E_m(k) \overset{\mathcal{D}}{\Longrightarrow} \max_{0\le t<\infty} \max_{0\le s\le t} w(t)\Big|W_1(t) - \frac{1+t}{1+s}W_1(s) + \frac{t-s}{1+s}W_2(1)\Big| \ , \tag{A.17}$$

and it only remains to investigate the impact of the covariance estimator. Therefore the following Lemma finishes the proof of Theorem 2.7.

Lemma A.4 (Plug in of covariance estimator) *We have that*

$$\sup_{k=1}^{\infty} w(k/m)E_m(k) - \sup_{k=1}^{\infty} w(k/m)\hat{E}_m(k) = o_{\mathbb{P}}(1) \ .$$

Proof. Observe the bound

$$\begin{aligned}
&\Big|\sup_{k=1}^{\infty} w(k/m)E_m(k) - \sup_{k=1}^{\infty} w(k/m)\hat{E}_m(k)\Big| \\
&\le \sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \max_{j=0}^{k-1}(k-j)\Big|(\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k})^\top(\hat{\Sigma}_m^{-1} - \Sigma^{-1})(\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k})\Big|^{1/2} \ .
\end{aligned} \tag{A.18}$$

Next note that for a symmetric matrix A and an arbitrary vector v the Cauchy-Schwarz inequality implies

$$|v^\top A v|^2 \le |Av||v| \le \|A\|_{op}|v|^2 \ ,$$

and we can bound (A.18) by

$$\Big\|\hat{\Sigma}_m^{-1} - \Sigma^{-1}\Big\|_{op}^{1/2} \sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \max_{j=0}^{k-1}(k-j)\Big|\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k}\Big| \ . \tag{A.19}$$

Since $\hat{\Sigma}_m$ is a consistent estimator of Σ, an application of the continuous mapping theorem yields

$$\Big\|\hat{\Sigma}_m^{-1} - \Sigma^{-1}\Big\|_{op} = o_{\mathbb{P}}(1) \ . \tag{A.20}$$

Next, the definition of the operator norm yields

$$\begin{aligned}
&\sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \max_{j=0}^{k-1}(k-j)\big|\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k}\big| \\
&\quad \le \|\Sigma^{1/2}\|_{op} \sup_{k=1}^{\infty} \frac{w(k/m)}{\sqrt{m}} \max_{j=0}^{k-1}(k-j)\big\|\hat{\theta}_1^{m+j} - \hat{\theta}_{m+j+1}^{m+k}\big\|_{\Sigma^{-1}} \\
&\quad = \|\Sigma^{1/2}\|_{op} \sup_{k=1}^{\infty} w(k/m)E_m(k) = \mathcal{O}_{\mathbb{P}}(1) \ .
\end{aligned}$$

Now a combination of (A.17) and (A.20) implies that the expression in (A.19) is of order $o_{\mathbb{P}}(1)$, which completes the proof of Lemma A.4 and thus also the proof of Theorem 2.7.

Combining Lemmas A.1, A.2, A.3 and A.4 we have now established that

$$\sup_{k=1}^{\infty} w(k/m)\hat{E}_m(k) \overset{\mathcal{D}}{\Longrightarrow} \sup_{0\le t<\infty} \max_{0\le s\le t} w(t)\left|W_1(t) - \frac{1+t}{1+s}W_1(s) + \frac{t-s}{1+s}W_2(1)\right|$$

and it remains to show that the distribution on the right-hand side of the last display is identical to the distribution on the right-hand side of (2.12).

Lemma A.5 (Simplify limit distribution) *It holds that*

$$\begin{aligned}\sup_{0\le t<\infty} \max_{0\le s\le t} w(t)&\left|W_1(t) - \frac{1+t}{1+s}W_1(s) + \frac{t-s}{1+s}W_2(1)\right| \\ &\overset{\mathcal{D}}{=} \sup_{0\le t<\infty} \max_{0\le s\le t} (t+1)w(t)\left|W\Big(\frac{t}{t+1}\Big) - W\Big(\frac{s}{s+1}\Big)\right| ,\end{aligned}$$

where W is a standard p-dimensional Brownian motion.

Proof of Lemma A.5 and last step in the proof of Theorem 2.7.
In the following let Z denote a vector of p independent standard Gaussian random variable, that is independent of W_1. Observe that

$$\begin{aligned}&w(t)\left|W_1(t) - \frac{1+t}{1+s}W_1(s) + \frac{t-s}{1+s}W_2(1)\right| \\ &= \frac{w(t)}{(s+1)}\Big|(s+1)W_1(t) - (t+1)W_1(s) + (t-s)W_2(1)\Big| \\ &\overset{\mathcal{D}}{=} \frac{w(t)}{(s+1)}\Big|(s+1)W(t) - (t+1)W(s) - (t-s)Z\Big| \\ &= \frac{w(t)}{(s+1)}\Big|(s+1)W(t) - (t+1)W(s) - (t-s)Z + stZ - stZ\Big| \\ &= \frac{w(t)}{(s+1)}\Big|(s+1)\big\{W(t) - tZ\big\} - (t+1)\big\{W(s) - sZ\big\}\Big| . \end{aligned} \tag{A.21}$$

Following Horváth et al. (2004), Fremdt (2015), computing the covariance function implies the following identity (in distribution)

$$\Big\{W(t) - tZ\Big\}_{t\ge 0} \overset{\mathcal{D}}{=} \Big\{(1+t)W\Big(\frac{t}{t+1}\Big)\Big\}_{t\ge 0} .$$

Applying this to (A.21) yields

$$\begin{aligned}
&\frac{w(t)}{(s+1)}\Big|(t+1)\{W(s)-sZ\}-(s+1)\{W(t)-tZ\}\Big| \\
\overset{\mathcal{D}}{=}&\frac{w(t)}{(s+1)}\Big|(t+1)(s+1)W\Big(\frac{s}{s+1}\Big)-(t+1)(s+1)W\Big(\frac{t}{t+1}\Big)\Big| \\
=&w(t)\Big|(t+1)W\Big(\frac{s}{s+1}\Big)-(t+1)W\Big(\frac{t}{t+1}\Big)\Big| \ .
\end{aligned}$$

This completes the proof of Lemma A.5 and also of Theorem 2.7.

□

Proof of Corollary 2.9: We proceed according to the proof of Theorem 3.1 of Fremdt (2015). Using the definition $w_\gamma(t) = \Big[(1+t)\max\Big\{\Big(\frac{t}{1+t}\Big)^\gamma, \varepsilon\Big\}\Big]^{-1}$, we obtain that

$$\begin{aligned}
&\sup_{0\le t<\infty}\max_{0\le s\le t} w_\gamma(t)\Big|(t+1)W\Big(\frac{s}{s+1}\Big)-(t+1)W\Big(\frac{t}{t+1}\Big)\Big| \\
&\quad= \sup_{0\le t<\infty}\max_{0\le s\le t}\frac{1}{\max\Big\{\Big(\frac{t}{1+t}\Big)^\gamma, \varepsilon\Big\}}\Big|W\Big(\frac{s}{s+1}\Big)-W\Big(\frac{t}{t+1}\Big)\Big| \\
&\quad= \sup_{0\le t<1}\max_{0\le s\le t}\frac{1}{\max\{t^\gamma,\varepsilon\}}\Big|W(s)-W(t)\Big| \ ,
\end{aligned}$$

where we used that the mapping $x \mapsto x/(1+x)$ is bijective and increasing on the domain $[0,\infty)$ with co-domain $[0,1)$. □

Proof of Theorem 2.13: For the ease of reading assume in the proof that $c_a m,\ c_a k_m^* \in \mathbb{N}$. We follow the idea of Stöhr (2019) and distinct the cases $k_m^*/m = O(1)$ and $k_m^*/m \to \infty$. In the first case, observe the lower bounds

$$\begin{aligned}
\sup_{k=1}^{\infty} w(k/m)\hat{E}_m(k) \ &= \sup_{k=1}^{\infty}\frac{w(k/m)}{\sqrt{m}}\max_{j=0}^{k-1}(k-j)\Big\|\hat\theta_1^{m+j}-\hat\theta_{m+j+1}^{m+k}\Big\|_{\hat\Sigma_m^{-1}} \\
&\overset{k=c_am+k_m^*}{\ge} \frac{w\Big(\frac{k_m^*}{m}+c_a\Big)}{\sqrt{m}}\max_{j=0}^{c_am+k_m^*-1}(c_am+k_m^*-j)\Big\|\hat\theta_1^{m+j}-\hat\theta_{m+j+1}^{(1+c_a)m+k_m^*}\Big\|_{\hat\Sigma_m^{-1}} \ .
\end{aligned}$$

Note that by Assumption 2.4 and 2.11 it holds that $w\Big(\frac{k_m^*}{m}+c_a\Big) = \tilde w\Big(\frac{k_m^*}{m}+c_a\Big)$ since $\frac{k_m^*}{m}+c_a \in (t_w, T_w)$. Thereby, the last display equals

$$\begin{aligned}
&\frac{\tilde w\Big(\frac{k_m^*}{m}+c_a\Big)}{\sqrt{m}}\max_{j=0}^{c_am+k_m^*-1}(c_am+k_m^*-j)\Big\|\hat\theta_1^{m+j}-\hat\theta_{m+j+1}^{(1+c_a)m+k_m^*}\Big\|_{\hat\Sigma_m^{-1}} \\
&\overset{j=k_m^*-1}{\ge} \frac{(c_am+1)\tilde w\Big(\frac{k_m^*}{m}+c_a\Big)}{\sqrt{m}}\Big\|\hat\theta_1^{m+k_m^*-1}-\hat\theta_{m+k_m^*}^{(1+c_a)m+k_m^*}\Big\|_{\hat\Sigma_m^{-1}} \ .
\end{aligned}$$

Using the reverse triangle inequality, the last display is bounded from below by

$$c_a \tilde{w}\Big(\tfrac{k_m^*}{m} + c_a\Big) \cdot \sqrt{m}\Bigg(\Big\|\theta_m^{(1)} - \theta_m^{(2)}\Big\|_{\hat{\Sigma}_m^{-1}} - \Big\|\hat{\theta}_1^{m+k_m^*-1} - \theta_m^{(1)}\Big\|_{\hat{\Sigma}_m^{-1}} - \Big\|\hat{\theta}_{m+k_m^*}^{(1+c_a)m+k_m^*} - \theta_m^{(2)}\Big\|_{\hat{\Sigma}_m^{-1}}\Bigg) . \tag{A.22}$$

To examine the first factor of the last display, note that we have $k_m^*/m \leq C$ for all $m \in \mathbb{N}$ and a sufficiently large constant C. Using Assumption 2.4, we now obtain

$$c_a \tilde{w}\Big(\tfrac{k_m^*}{m} + c_a\Big) \geq c_a \min_{t \in [c_a, C+c_a]} \tilde{w}(t) > 0 .$$

Now it remains to treat the second factor in (A.22). Note that by Assumption 2.11, we obtain that

$$\sqrt{m}\Big|\theta_m^{(1)} - \theta_m^{(2)}\Big| \underset{m\to\infty}{\Longrightarrow} \infty .$$

Using also the linearization in equation (2.7) and (2.17), we conclude that

$$\sqrt{m}\Big(\hat{\theta}_1^{m+k_m^*-1} - \theta_m^{(1)}\Big) = \frac{\sqrt{m}}{m + k_m^* - 1} \sum_{t=1}^{m+k_m^*-1} \mathcal{IF}_t \; + \; \sqrt{m} R_{1,m+k_m^*-1} = \mathcal{O}_{\mathbb{P}}(1)$$

and

$$\sqrt{m}\Big(\hat{\theta}_{m+k_m^*}^{(1+c_a)m+k_m^*} - \theta_m^{(2)}\Big) = \frac{\sqrt{m}}{c_a m + 1} \sum_{t=m+k_m^*}^{(1+c_a)m+k_m^*} \mathcal{IF}_t \; + \; \sqrt{m} R_{m+k^*,(1+c_a)m+k_m^*} = \mathcal{O}_{\mathbb{P}}(1) .$$

Putting all together and using also that $\hat{\Sigma}_m$ is (weakly) convergent with non-singular limit the treatment of the first case is finished since (A.22) diverges to ∞.

It remains to treat the case $k_m^*/m \to \infty$, for which we can employ very similar arguments. Setting $k = k_m^*(1 + c_a)$ and $j = k_m^* - 1$ in the definition of $w(k/m)\hat{E}_m(k)$ gives the lower bound

$$\sup_{k=1}^{\infty} w(k/m)\hat{E}_m(k) \geq \frac{c_a k_m^* w\Big(\frac{k_m^*}{m}(1 + c_a)\Big)}{\sqrt{m}} \Big\|\hat{\theta}_1^{m+k_m^*-1} - \hat{\theta}_{m+k_m^*}^{m+(1+c_a)k_m^*}\Big\|_{\hat{\Sigma}_m^{-1}} . \tag{A.23}$$

As $T_w = \infty$ by assumption, we have $w\Big(\frac{k_m^*}{m}(1 + c_a)\Big) = \tilde{w}\Big(\frac{k_m^*}{m}(1 + c_a)\Big)$ for m sufficiently

large. Now we obtain that (A.23) has the lower bound

$$
\begin{aligned}
&c_a \frac{k_m^*}{m} \tilde{w}\Big(\tfrac{k_m^*}{m}(1+c_a)\Big) \\
&\quad \cdot \sqrt{m}\left(\left\| \theta_m^{(1)} - \theta_m^{(2)} \right\|_{\hat{\Sigma}_m^{-1}} - \left\| \hat{\theta}_1^{m+k_m^*-1} - \theta_m^{(1)} \right\|_{\hat{\Sigma}_m^{-1}} - \left\| \hat{\theta}_{m+k_m^*}^{m+(1+c_a)k_m^*} - \theta_m^{(2)} \right\|_{\hat{\Sigma}_m^{-1}} \right) .
\end{aligned}
\tag{A.24}
$$

By assumption (2.19) we obtain

$$
\liminf_{m\to\infty} \frac{k_m^*}{m} \tilde{w}\Big(\tfrac{k_m^*}{m}(1+c_a)\Big) > 0 .
$$

Using (2.18) and repeating the corresponding steps from the first case, it follows that

$$
\sqrt{m}\left(\left\| \theta_m^{(1)} - \theta_m^{(2)} \right\|_{\hat{\Sigma}_m^{-1}} - \left\| \hat{\theta}_1^{m+k_m^*-1} - \theta_m^{(1)} \right\|_{\hat{\Sigma}_m^{-1}} - \left\| \hat{\theta}_{m+k_m^*}^{m+(1+c_a)k_m^*} - \theta_m^{(2)} \right\|_{\hat{\Sigma}_m^{-1}} \right) \overset{\mathbb{P}}{\Longrightarrow} \infty .
$$

Combining the last two statements with the lower bound provided in (A.24) the treatment of the second case and thereby the proof of Theorem 2.13 is finished. □

B Additional simulation results for Section 4

In this section we provide some additional simulation results complementing the discussion on the power of the different monitoring procedures in Section 4. The simulation settings are identical to those used in Section 4 and as the results below are very similar to the results displayed in Figures 1 and 2 we omit a further discussion here.

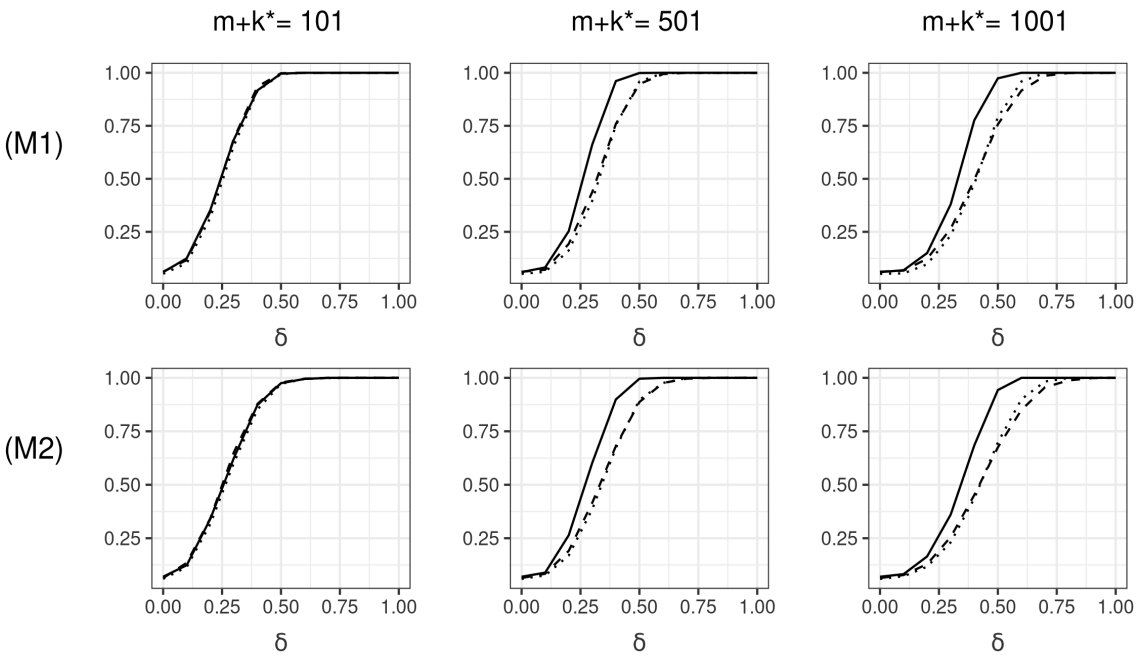

Figure 6: *Power of the monitoring procedures for a change in the mean based on the statistics $\hat{E}$ (solid line), $\hat{Q}$ (dashed line) and $\hat{P}$ (dotted line) with $\gamma = 0.45$ and $m = 100$.*

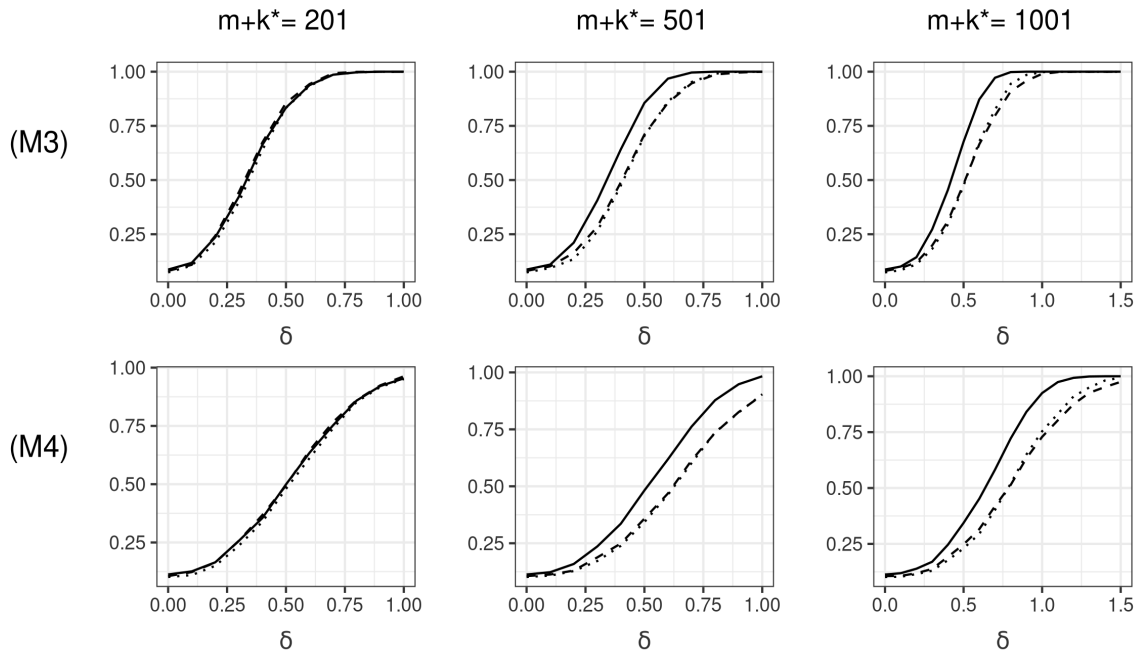

Figure 7: *Power of the monitoring procedures for a change in the mean based on the statistics $\hat{E}$ (solid line), $\hat{Q}$ (dashed line) and $\hat{P}$ (dotted line) with $\gamma = 0.45$ and $m = 200$.*

C Closed-end scenarios

It is worthwhile to mention that the theory developed in Section 2 also covers the case of closed-end scenarios [sometimes also called finite time horizon]. In this section, we will very briefly discuss this situation and present a small batch of simulation results, which also indicate the superiority of the statistic $\hat{E}$ for closed-end scenarios. Note that the null hypothesis in this setup is given by

$$H_0 : \ \theta_1 = \cdots = \theta_m = \theta_{m+1} = \theta_{m+2} = \ldots = \theta_{(T+1)m} \ , \tag{C.1}$$

which is tested against the alternative that the parameters changes (once) at some time $m + 1 \leq m + k^\star \leq (T + 1)m$, that is

$$H_1 : \ \exists k^\star \in \mathbb{N} : \ \ \theta_1 = \cdots = \theta_{m+k^\star-1} \neq \theta_{m+k^\star} = \theta_{m+k^\star+1} = \ldots = \theta_{(T+1)m} \ . \tag{C.2}$$

Here the factor $T \in \mathbb{N}$ controls the length of the monitoring period compared to the size of the initial data set. Under the assumptions stated in Section 2, we can prove a corresponding statement of Theorem 2.7 and Corollary 2.9.

Theorem C.1 *Assume that the null hypothesis* (C.1) *and Assumptions 2.3 - 2.5 hold. If further $\hat{\Sigma}_m$ is a consistent and non-singular estimator of the long-run variance matrix Σ it*

holds that

$$\sup_{k=1}^{Tm} w_\gamma(k/m)\hat{E}_m(k) \overset{\mathcal{D}}{\Longrightarrow} \sup_{0\le t\le T}\max_{0\le s\le t}(t+1)w(t)\Big|W\Big(\frac{t}{t+1}\Big) - W\Big(\frac{s}{s+1}\Big)\Big| \tag{C.3}$$

where W is a p-dimensional Brownian motion with independent components. Using $w = w_\gamma$ for the class of weight functions defined in (2.14), we further obtain that

$$\begin{aligned}&\sup_{0\le t\le T}\max_{0\le s\le t}(t+1)w_\gamma(t)\Big|W\Big(\frac{t}{t+1}\Big) - W\Big(\frac{s}{s+1}\Big)\Big| \\ &\qquad \overset{\mathcal{D}}{=} \max_{0<t\le T/(T+1)}\max_{0\le s\le t}\frac{1}{\max\{t^\gamma,\varepsilon\}}\Big|W(t)-W(s)\Big| =: L_{1,\gamma}(T)\ .\end{aligned} \tag{C.4}$$

The proof of Theorem C.1 follows from Theorem 2.7 by using the factor T as the cutoff T_w of the weight function in (2.10). The representation provided in (C.4) follows from a straightforward adaption of Corollary 2.9. The corresponding results for the tests based on statistics $\hat{Q}$ and $\hat{P}$ defined in (4.1) read as follows

$$\max_{k=1}^{Tm} w_\gamma(k/m)\hat{Q}_m(k) \overset{\mathcal{D}}{\Longrightarrow} \max_{0<t<T/(T+1)}\frac{|W(t)|}{\max\{t^\gamma,\varepsilon\}} =: L_{2,\gamma}(T) \tag{C.5}$$

and

$$\max_{k=1}^{Tm} w_\gamma(k/m)\hat{P}_m(k) \overset{\mathcal{D}}{\Longrightarrow} \max_{0<t<T/(T+1)}\max_{0\le s\le t}\frac{1}{\max\{t^\gamma,\varepsilon\}}\Big|W(t) - \frac{1-t}{1-s}W(s)\Big| =: L_{3,\gamma}(T)\ . \tag{C.6}$$

Likewise to Remark 2.10 we can obtain an exact formula for the distribution of $L_{1,0}(T)$ in the case $p = 1$ from page 146 of Borodin and Salminen (1996), this is

$$F_{L_1(T),\gamma=0}(x) = 1 + 8\sum_{k=1}^{\infty}(-1)^k\cdot k\cdot\Big(1-\Phi\big(kx/\sqrt{q(T)}\big)\Big)\ , \tag{C.7}$$

where Φ denotes the c.d.f. of a standard Gaussian random variable and $q(T)$ denotes the quotient $T/(T+1)$.

To complete the discussion on closed-end scenarios we will display a small batch of simulation results for the detection of changes in the mean as described in Section 3.1. For the sake of brevity, only the choice $T = 4$ is examined here [unpublished simulation results show similar outcomes for other choices of T]. The remaining simulation settings are the same as used for the simulation study presented in Section 4.1 and in Table 5 we display the necessary critical values defining the rejection regions for the different procedures.

The approximation of the nominal level under the null hypothesis is displayed in Tables 6 and 7 and in Figures 8 and 9 the power of the different procedures with respect to change

amount and change position for $\gamma = 0$ is illustrated. The results are very similar to the open-end scenario discussed in Section 4 and confirm the findings of that Section.

	$L_{1,\gamma}(4)$			$L_{2,\gamma}(4)$			$L_{3,\gamma}(4)$		
$\gamma \setminus \alpha$	0.01	0.05	0.1	0.01	0.05	0.1	0.01	0.05	0.1
0	2.7042	2.2339	2.0046	2.5145	1.9826	1.7380	2.5572	2.0435	1.8019
0.25	2.9558	2.4345	2.2220	2.7602	2.2223	1.9799	2.8210	2.2986	2.0750
0.45	3.3850	2.9371	2.6994	3.2238	2.7398	2.4952	3.3156	2.8626	2.6274

Table 5: *(1-α)-quantiles of the distributions $L_{1,\gamma}(4)$, $L_{2,\gamma}(4)$ and $L_{3,\gamma}(4)$ for different choices of γ. The cutoff constant was set to $\varepsilon = 0$ and the dimension is $p = 1$. The quantiles for $L_{1,0}(4)$ were computed with respect to formula (C.7).*

	(M1)			(M2)		
γ	$\hat{E}$	$\hat{Q}$	$\hat{P}$	$\hat{E}$	$\hat{Q}$	$\hat{P}$
0	5.0%	5.3%	5.3%	8.0%	7.3%	7.4%
0.25	5.4%	6.0%	5.8%	8.6%	7.5%	8.1%
0.45	4.9%	5.4%	4.5%	6.1%	6.4%	5.9%

Table 6: Type I error for the closed-end procedures for a change in the mean based on the statistics $\hat{E}$, $\hat{Q}$ and $\hat{P}$ at 5% nominal size with a training data set of size $m = 200$ and a monitoring window of $T = 4$.

	(M3)			(M4)		
γ	$\hat{E}$	$\hat{Q}$	$\hat{P}$	$\hat{E}$	$\hat{Q}$	$\hat{P}$
0	7.6%	7.8%	8.1%	9.9%	9.8%	10.1%
0.25	8.4%	8.2%	8.7%	11.4%	10.4%	10.8%
0.45	6.6%	6.5%	6.8%	8.9%	8.3%	8.5%

Table 7: Type I error for the closed-end procedures for a change in the mean based on the statistics $\hat{E}$, $\hat{Q}$ and $\hat{P}$ at 5% nominal size with a training data set of size $m = 400$ and a monitoring window of $T = 4$.

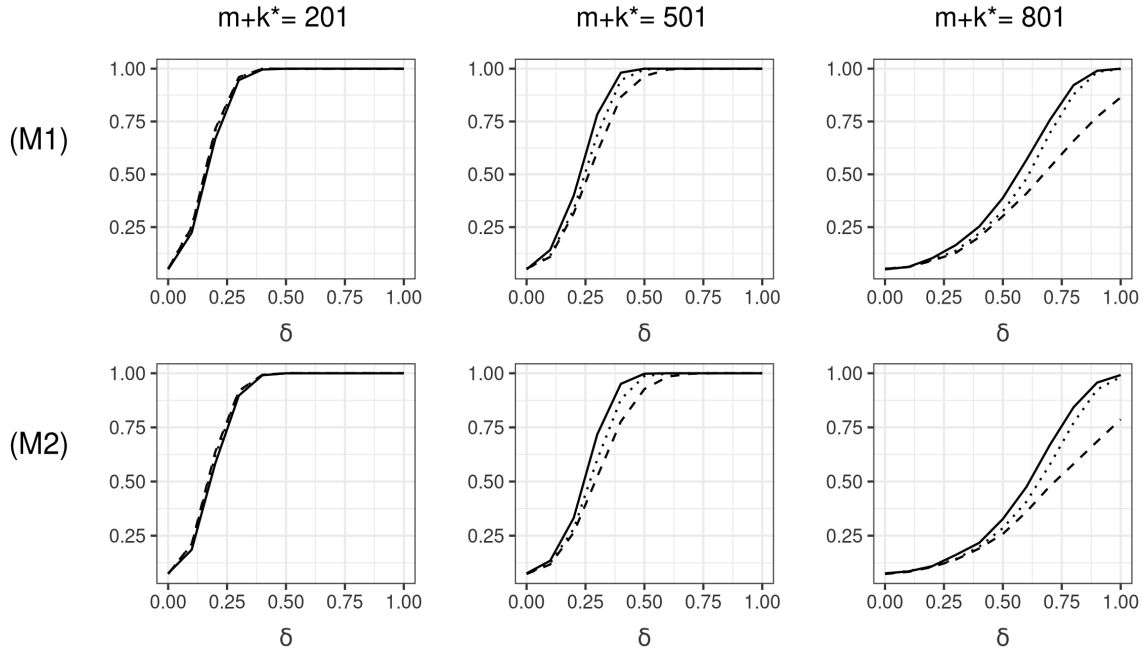

Figure 8: *Power of the (closed-end) monitoring procedures for a change in the mean based on the statistics* $\hat{E}$ *(solid line),* $\hat{Q}$ *(dashed line) and* $\hat{P}$ *(dotted line) with* $\gamma = 0$*,* $m = 200$ *and* $T = 4$*.*

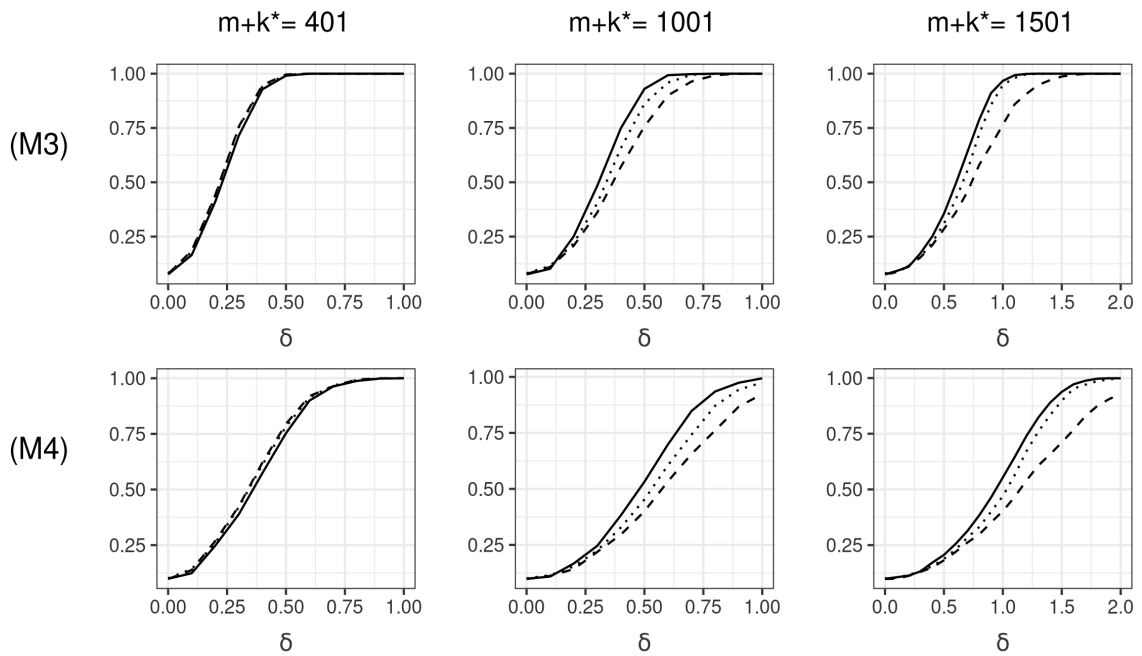

Figure 9: *Power of the (closed-end) monitoring procedures for a change in the mean based on the statistics* $\hat{E}$ *(solid line),* $\hat{Q}$ *(dashed line) and* $\hat{P}$ *(dotted line) with* $\gamma = 0$*,* $m = 400$ *and* $T = 4$*.*

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